We consider how to initialize the parameters of a model before we start training.

1. Why is proper initialization important?

During forward process:

$$f_k = \beta_k + \Omega_k h_k$$
$$= \beta_k + \Omega_k a(f_{k-1})$$

(a: activation function, e.g. ReLu)

if β_k is initialized to 0 and Ω_k is initialized such that each element follows $N(0, \sigma^2)$ independently.

Then if σ is small, e.g. 0.001, then the magnitude of $f_k \ll f_{k+1}$

if σ is large, e.g. 100, then the magnitude of $f_k \gg f_{k+1}$

in deep neural networks. this is even more serious, because the magnitude problem will be worse and worse through each layer.

During backpropagation:

$$\frac{\partial l}{\partial f_{k-1}} = \frac{\partial l}{\partial f_k}\frac{\partial f_k}{\partial f_{k-1}}$$
$$= \frac{\partial l}{\partial f_k}\Omega_k \frac{\partial (a(f_{k-1}))}{\partial f_{k-1}}$$
$$= \frac{\partial l}{\partial f_k}\Omega_k \, diag\left(\mathbb{I}(f_{k-1} \geq 0)\right) \quad \text{(if } a = \text{relu)}$$

Depends on the scale of Ω_k, there is risk (if we do not initialize properly) that the gradients → getting smaller and smaller → hard to train!
→ getting larger and larger → hard to train!

Solution: ideally we want to keep the variance unchanged during both in other words,

forward pass: $Var(f_k) \approx Var(f_{k-1})$

backward propagation: $Var(\frac{\partial l}{\partial f_{k-1}}) \approx Var(\frac{\partial l}{\partial f_k})$

Derivation: (β_k will be initialized to 0)

notations: $f: f_{k-1}$, $f': f_k$, f_i: i th element of f, f_i': i th element of f', $\beta: \beta_k$, $\Omega: \Omega_k$

1. forward pass:

$$E(f_i') = E(\beta_i + \sum_{j=1}^{m} \Omega_{ij} a(f_j))$$

$$= E(\beta_i) + E(\sum_{j=1}^{m} \Omega_{ij} a(f_j))$$

$$= 0 + \sum_{j=1}^{m} E(\Omega_{ij} a(f_j))$$

$$= \sum_{j=1}^{m} E(\Omega_{ij}) E(a(f_j))$$ the parameters are initialized independently wrt the layer outputs

$$= \sum_{j=1}^{m} 0 \cdot E(a(f_j))$$

$$= 0$$

$$Var(f_i') = E(f_i'^2) - E(f_i')^2$$

$$= E(f_i'^2) - 0$$

$$= E((\beta_i + \sum_{j=1}^{m} \Omega_{ij} a(f_j))^2)$$

$$= E(\sum_{j=1}^{m} (\Omega_{ij} a(f_j))^2)$$ ($E(\beta_i') = E(\Omega_{ij}) = 0$)

$$= \sum_{j=1}^{m} E(\Omega_{ij}^2) E(a(f_j)^2)$$ (independence)

$$= \sigma_\Omega^2 \sum_{j=1}^{m} E(a(f_j)^2)$$

$$= \frac{1}{2} m \sigma_\Omega^2 \sigma_f^2$$

assume $a(\cdot)$ is ReLU, then

$$E(a(f_j)^2) = \int_{-\infty}^{+\infty} \max(0, f_j)^2 p(f_j)\, df_j$$

$$= \int_0^{\infty} f_j^2 p(f_j)\, df_j$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} f_j^2 p(f_j)\, df_j$$ assume $p(f_j)$ is symmetric around 0

$$= \frac{1}{2} \sigma_{f_j}^2$$ $E(f_j) = 0$

Conclusion: if we want $\sigma_{f_l}^2 = \sigma_f^2$, we need to set

$$\sigma_\Omega^2 = \frac{2}{m} \quad \text{(He initialization)}$$

↘ number of neurons for last layer, i.e. rows for Ω_{k-1}

2. Backward propagation: (follows the same logic)

First, recall that:

$$\begin{aligned}\frac{\partial l}{\partial f_{k-1}} &= \frac{\partial l}{\partial f_k}\frac{\partial f_k}{\partial f_{k-1}} \\ &= \frac{\partial l}{\partial f_k}\Omega_k \frac{\partial (a(f_{k-1}))}{\partial f_{k-1}} \\ &= \frac{\partial l}{\partial f_k}\Omega_k \, diag\left(\mathbb{I}(f_{k-1}\geq 0)\right)\end{aligned}$$

for notation simplicity, let us denote $\frac{\partial l}{\partial f_k} = g'$ and $\frac{\partial l}{\partial f_{k-1}} = g$

$\Omega_k = \Omega$, $f = f_{k-1}$ and $f' = f_k$

So we have: $E(g_i) = E\left(\sum_{j=1}^{m'} g_j' \Omega_{ji} (I(f_i\geq 0))\right)$

$$\begin{aligned}&= \sum_{j=1}^{m'} E\left(g_j' \Omega_{ji} I(f_i\geq 0)\right) \\ &= \sum_{j=1}^{m'} E(\Omega_{ji}) E(g_j' I(f_i\geq 0)) \\ &= \sum_{j=1}^{m'} 0 \cdot E(g_j' I(f_i\geq 0)) \\ &= 0\end{aligned}$$

$$\begin{aligned}Var(g_i) = \sigma_g^2 &= E(g_i^2) - E(g_i)^2 \\ &= E(g_i^2) \\ &= E\left(\left(\sum_{j=1}^{m} g_i' \Omega_{ji} I(f_i\geq 0)\right)^2\right) \\ &= \sum_{j=1}^{m} E\left(g_i'^2 \Omega_{ji}^2 I(f_i\geq 0)^2\right) && E(\Omega_{ji}) = 0 \text{ and independence} \\ &= \sum_{j=1}^{m} \sigma_\Omega^2 E\left(g_i'^2 I(f_i\geq 0)^2\right) && \text{assume independence} \\ &= \frac{1}{2}\sum_{j=1}^{m} \sigma_\Omega^2 \sigma_{g'}^2\end{aligned}$$

Conclusion: if we want $\sigma_{g_l}^2 = \sigma_g^2$, we need $\sigma_\Omega^2 = \frac{2}{m'}$

If $m = m'$, then we can initialize Ω to keep both forward and backward stable

if $m \neq m'$, we use $\sigma_\Omega^2 = \frac{4}{m+m'}$ → replace m or m' with $\frac{m+m'}{2}$